

HSS SHANGUS

Govt Higher Secondary School Shangus

GENERAL KNOWLEDGE QUIZ

Complete Study Material

Classes 9th – 12th

All Streams — Science, Arts & General

Prepared by the Team of Subject Experts
under the guidance of the Principal, HSS Shangus

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About This Study Material

This booklet has been prepared as complete study material for the General Knowledge Quiz for students of Classes 9th to 12th, across all streams — Science, Arts and others. Every topic is written at a general-awareness level: the aim is knowledge that any student should reasonably have, not stream-specific depth (for example, no complex chemical reactions, legal analysis, or advanced algebra).

Each section follows the same pattern: a short explanation of the topic, key facts organised as bullet points or tables, and — where useful — a worked example. The Current Affairs section is necessarily time-sensitive; students are encouraged to refresh it using recent newspapers or news sources closer to the date of the quiz.

Total: 60 Marks across 60 Questions (1 mark each), distributed across the ten topics below. See the *Practice Questions* section near the end for sample questions, followed by the *Answer Key and Marks & Question Distribution* on the final page.

1. Indian Freedom Struggle (1857–1947)

India's journey to independence spanned nearly a century of resistance against British colonial rule — from the first large-scale uprising in 1857 to the transfer of power in August 1947. This section covers the major milestones most commonly tested in general knowledge quizzes.

Revolt of 1857

Also called the First War of Independence (or the Sepoy Mutiny by British historians), this was the first large-scale armed uprising against East India Company rule.

- **Causes:** political annexations under the Doctrine of Lapse (Lord Dalhousie), economic exploitation and ruin of traditional industries, interference in social and religious customs, and discontent among Indian soldiers (sepoys) — sharpened by the rumour that new cartridges were greased with cow and pig fat.
- **Spark:** Mangal Pandey's revolt at Barrackpore (March 1857), followed by a wider outbreak at Meerut on 10 May 1857.
- **Key leaders:** Rani Lakshmibai (Jhansi), Tantia Tope and Nana Sahib (Kanpur), Bahadur Shah Zafar (symbolic leader, Delhi), and Begum Hazrat Mahal (Awadh).
- **Consequences:** East India Company rule ended; the Government of India Act, 1858 transferred power directly to the British Crown.

Formation of the Indian National Congress (1885)

The Indian National Congress (INC) was founded in 1885 by Allan Octavian Hume, a retired British civil servant, with its first session held in Bombay. In its early ('Moderate') phase it sought reform through constitutional and petition-based methods; it later became the principal organisation leading the freedom movement.

Partition of Bengal (1905)

Ordered by Viceroy Lord Curzon, officially for administrative convenience but widely seen as an attempt to weaken nationalist unity along religious lines. It triggered the Swadeshi Movement — a boycott of British goods and promotion of Indian industries — and was annulled in 1911 following sustained public protest.

Home Rule Movement (1916)

Inspired by the Irish Home Rule movement, this campaign was led by Bal Gangadhar Tilak and Annie Besant through two parallel Home Rule Leagues, demanding self-government for India within the British Empire.

Jallianwala Bagh Massacre (13 April 1919)

On the festival of Baisakhi, a peaceful public gathering at Jallianwala Bagh in Amritsar was fired upon on the orders of General Reginald Dyer, killing and wounding hundreds. It became one of the most significant turning points that hardened Indian public opinion against British rule.

Non-Cooperation Movement (1920–1922)

Led by Mahatma Gandhi, this movement called for the boycott of British goods, institutions, titles and legislative councils. It was withdrawn after the Chauri Chaura incident (February 1922), in which a violent clash led to the deaths of policemen.

Civil Disobedience Movement & Dandi March (1930)

The movement began with the Dandi March (Salt March): Gandhi and his followers walked about 390 km from Sabarmati Ashram to Dandi over 24 days to break the British salt law by making salt from seawater. It triggered a nationwide campaign of civil disobedience against unjust laws.

Quit India Movement (1942)

Launched by Gandhi with the call 'Do or Die,' this movement demanded an immediate end to British rule. It was met with the mass arrest of leaders, and continued as a largely spontaneous, leaderless protest — one of the last major mass movements before independence on 15 August 1947.

Quick Timeline

Year	Event
1857	Revolt of 1857 (First War of Independence)
1885	Formation of the Indian National Congress
1905	Partition of Bengal; Swadeshi Movement begins
1916	Home Rule Movement
1919	Jallianwala Bagh Massacre
1920–22	Non-Cooperation Movement
1930	Civil Disobedience Movement; Dandi March
1942	Quit India Movement
1947	India attains Independence (15 August)

Key Personalities of the Freedom Struggle

Personality	Known For
Mahatma Gandhi	Father of the Nation; philosophy of Satyagraha (non-violent resistance)
Jawaharlal Nehru	First Prime Minister of India
Dr. B.R. Ambedkar	Chief architect of the Indian Constitution
Sardar Vallabhbhai Patel	'Iron Man of India'; integrated the princely states into the Union
Subhas Chandra Bose	Founder of the Indian National Army (INA); known as 'Netaji'
Bhagat Singh	Revolutionary; gave the slogan 'Inquilab Zindabad'
Sarojini Naidu	'Nightingale of India'; first woman President of the INC
Rani Lakshmibai	Led resistance during the Revolt of 1857 from Jhansi

Famous Slogans of the Freedom Struggle

Slogan	Given By
'Swaraj is my birthright, and I shall have it'	Bal Gangadhar Tilak
'Inquilab Zindabad' (Long Live the Revolution)	Bhagat Singh
'Give me blood, and I will give you freedom'	Subhas Chandra Bose
'Delhi Chalo'	Subhas Chandra Bose (INA)
'Do or Die'	Mahatma Gandhi (Quit India Movement)
'Jai Hind'	Subhas Chandra Bose
'Vande Mataram'	Bankim Chandra Chattopadhyay (from Anandmath)

2. Political Science: Constitution & Governance

The Constitution of India was adopted by the Constituent Assembly on 26 November 1949 and came into force on 26 January 1950 — celebrated since as Republic Day. Dr. B.R. Ambedkar chaired the Drafting Committee, and it remains the longest written constitution of any sovereign country in the world.

Fundamental Rights (Articles 12–35)

Fundamental Rights are basic guarantees available to every citizen, enforceable in a court of law. They are grouped into six categories:

Right	Articles
Right to Equality	14–18
Right to Freedom	19–22
Right against Exploitation	23–24
Right to Freedom of Religion	25–28
Cultural and Educational Rights	29–30
Right to Constitutional Remedies	32

Note: Article 32 is often called the 'heart and soul' of the Constitution — a phrase used by Dr. Ambedkar — because it allows citizens to approach the Supreme Court directly if their Fundamental Rights are violated. The Right to Property was originally a Fundamental Right but was made a legal right under Article 300-A by the 44th Amendment (1978).

Fundamental Duties (Article 51A)

Added by the 42nd Amendment (1976) on the recommendation of the Swaran Singh Committee, these are moral obligations of every citizen. Originally 10 duties were listed; an 11th was added by the 86th Amendment (2002), requiring parents/guardians to provide education to children aged 6–14. Representative duties include:

- Respect the Constitution, the National Flag and the National Anthem.
- Cherish and follow the noble ideals of the freedom struggle.
- Uphold and protect the sovereignty, unity and integrity of India.
- Protect and improve the natural environment, including forests and wildlife.
- Develop scientific temper, humanism and the spirit of inquiry and reform.
- Safeguard public property and abjure violence.
- Strive towards excellence in all spheres of individual and collective activity.

Landmark Articles (illustrative, not exhaustive)

Article	Subject
14	Equality before law and equal protection of laws
19	Six freedoms — speech, assembly, association, movement, residence, profession
21	Right to life and personal liberty
32	Right to Constitutional Remedies
44	Uniform Civil Code (a Directive Principle, not a Fundamental Right)
356	President's Rule in a state
370	Special provisions for Jammu & Kashmir (see note below)

Note: Article 370 originally granted special autonomous status to the former state of Jammu & Kashmir. In August 2019, Parliament revoked its operative provisions, and with effect from 31 October 2019, J&K was reorganised into two Union Territories — Jammu & Kashmir and Ladakh.

Structure of Government

- **Legislature (Parliament):** Lok Sabha (House of the People, directly elected) and Rajya Sabha (Council of States, representing the states).

- **Executive:** The President (constitutional head of state), the Prime Minister (head of government) and the Council of Ministers.
- **Judiciary:** The Supreme Court (apex court), High Courts, and subordinate courts — independent of the legislature and executive.

Political Parties

The Election Commission of India (ECI) classifies parties as National Parties, State (regional) Parties, or Registered Unrecognised Parties, based on electoral performance and presence.

- As of 2026, the ECI recognises six National Parties: Bharatiya Janata Party (BJP), Indian National Congress (INC), Bahujan Samaj Party (BSP), Communist Party of India (Marxist) [CPI(M)], Aam Aadmi Party (AAP), and National People's Party (NPP).
- Numerous State/regional parties exist as well — for example DMK, TMC, Shiv Sena, RJD, Samajwadi Party (SP), YSRCP and BJD.

Note: Party recognition status is reviewed by the ECI after elections and can change. Re-verify this list closer to the date of the quiz.

National Symbols of India

Symbol	Details
National Flag	The Tiranga — saffron, white and green with the Ashoka Chakra (24 spokes)
National Emblem	Lion Capital of Ashoka (from Sarnath); motto 'Satyameva Jayate' (Truth alone triumphs)
National Anthem	Jana Gana Mana — written by Rabindranath Tagore
National Song	Vande Mataram — written by Bankim Chandra Chattopadhyay
National Animal	Royal Bengal Tiger
National Bird	Peacock
National Flower	Lotus
National Fruit	Mango
National Tree	Banyan

Important Constitutional & Statutory Bodies

- **Election Commission of India (ECI):** conducts free and fair elections (Article 324).
- **Union Public Service Commission (UPSC):** recruits for the All India and Central civil services (Article 315).
- **Comptroller and Auditor General (CAG):** audits the accounts of the Union and state governments (Article 148).
- **Finance Commission:** recommends how tax revenues are distributed between the Centre and the states (Article 280).
- **NITI Aayog:** the government's policy think tank, which replaced the Planning Commission in 2015 (a statutory, not constitutional, body).

3. Environmental Studies

Environmental Studies looks at how human activity affects the planet's natural systems, and what can be done to protect them. This section covers four core themes commonly tested in quizzes.

Greenhouse Effect & Global Warming

The greenhouse effect is a natural process in which certain gases — carbon dioxide, methane, water vapour and nitrous oxide — trap heat in the atmosphere, keeping Earth warm enough to sustain life. Human activities such as burning fossil fuels, deforestation and industrial processes have intensified this effect, causing global warming: a long-term rise in Earth's average temperature.

- **Consequences:** melting glaciers and polar ice, rising sea levels, more frequent extreme weather events, and threats to biodiversity and agriculture.
- **Kyoto Protocol (1997):** an international treaty under the UN Framework Convention on Climate Change that committed industrialised countries to binding targets for reducing greenhouse gas emissions.

Ozone Layer Depletion

The ozone layer, found in the stratosphere, absorbs most of the sun's harmful ultraviolet (UV) radiation. Its depletion is mainly caused by chlorofluorocarbons (CFCs) and related chemicals, once widely used in refrigerants, aerosols and foam products.

- **Consequences:** increased UV radiation reaching Earth's surface, a higher risk of skin cancer and cataracts, and harm to marine ecosystems.
- **Montreal Protocol (1987):** an international agreement to phase out ozone-depleting substances — widely regarded as one of the most successful environmental treaties ever adopted.

Greenhouse Effect vs Ozone Depletion — At a Glance

Aspect	Greenhouse Effect	Ozone Depletion
Main gases involved	CO ₂ , methane, water vapour, N ₂ O	Chlorofluorocarbons (CFCs)
Main consequence	Global warming, climate change	Increased UV radiation reaching Earth
Related treaty	Kyoto Protocol (1997)	Montreal Protocol (1987)

Biodiversity

Biodiversity refers to the variety of life on Earth across three levels — species, genetic, and ecosystem diversity. It is important for ecological balance, food security, medicine and climate regulation. Major threats include habitat destruction, pollution, climate change, overexploitation of resources, and invasive species.

Renewable & Non-Renewable Resources

- **Renewable resources:** naturally replenished resources such as solar, wind, hydro, biomass and geothermal energy.
- **Non-renewable resources:** finite resources that take millions of years to form, such as coal, petroleum, natural gas and minerals.

Sustainable development depends on balancing the use of both categories, with a growing global shift toward renewable sources of energy.

Environmental Movements in India

The Chipko Movement (1973, Uttarakhand) was a grassroots forest-conservation movement in which villagers — including Sunderlal Bahuguna and Gaura Devi — hugged trees to prevent them from being felled by loggers. It remains one of India's most iconic environmental movements and inspired similar efforts worldwide.

Important Environmental Days

Day	Date
World Wildlife Day	3 March

Day	Date
World Water Day	22 March
Earth Day	22 April
World Environment Day	5 June
World Ozone Day	16 September

Key International Environmental Agreements

- **Ramsar Convention (1971):** an international treaty for the conservation of wetlands.
- **Convention on Biological Diversity (CBD, 1992):** a global agreement for the conservation of biodiversity and sustainable use of its components.
- **Paris Agreement (2015):** a landmark global pact under the UNFCCC to limit global warming, building on the Kyoto Protocol.

Types of Pollution

Type	Common Causes
Air Pollution	Vehicle emissions, industrial smoke, burning of fossil fuels
Water Pollution	Industrial effluents, sewage discharge, agricultural runoff
Soil Pollution	Pesticides, chemical fertilisers, improper waste disposal
Noise Pollution	Traffic, industrial machinery, loudspeakers

The 3 R's of Conservation

- **Reduce:** use fewer resources and generate less waste in the first place.
- **Reuse:** use items again rather than discarding them after a single use.
- **Recycle:** process used materials (paper, plastic, metal, glass) into new products.

4. Basic Economics & Banking

Banking and financial literacy are essential everyday-life knowledge for students of every stream. This section covers banks, the RBI, account types, payment instruments and digital payments.

Banks — Meaning, Functions & Types

A bank is a financial institution that accepts deposits from the public and lends money for productive purposes. Its core functions include accepting deposits, lending loans and advances, facilitating payments, providing safekeeping of valuables (lockers), and agency services.

- **Central Bank:** Reserve Bank of India (RBI) — regulates the entire banking system.
- **Commercial Banks:** Public Sector Banks (e.g. SBI, PNB), Private Sector Banks (e.g. HDFC, ICICI), and Foreign Banks.
- **Other categories:** Regional Rural Banks, Cooperative Banks, Small Finance Banks, and Payments Banks.

Reserve Bank of India (RBI)

India's central bank was established on 1 April 1935 under the RBI Act, 1934, and is headquartered in Mumbai. Its key functions include issuing currency notes, regulating money supply and interest rates through monetary policy, acting as banker to the government and to other banks, maintaining financial stability, and managing the country's foreign exchange reserves.

Note: As of 2026, the RBI Governor is Sanjay Malhotra (26th Governor). Office-holders change periodically — re-verify before the quiz.

Types of Bank Accounts

Account Type	Key Feature
Savings Account	For individuals; encourages saving; limited transactions; earns interest
Current Account	For businesses; unlimited transactions; usually no interest
Fixed Deposit (FD)	Lump sum deposited for a fixed period at a fixed interest rate
Recurring Deposit (RD)	Fixed amount deposited every month for a fixed period

Payment Instruments

- **Cheque:** a written order instructing a bank to pay a specified amount.
- **Passbook:** a record of all transactions in a savings account.
- **Debit Card:** linked directly to a bank account; money is deducted immediately.
- **Credit Card:** allows borrowing up to a limit, repaid later (often with interest if unpaid on time).

Digital Payment Systems

System	How It Works
UPI (Unified Payments Interface)	Instant, real-time payments between bank accounts using a mobile app (Phonepe, BHIM, GPay); used for daily transactions
NEFT (National Electronic Funds Transfer)	Batch-based fund transfer, settled in half-hourly cycles; best for bank to bank transfers
RTGS (Real Time Gross Settlement)	Real-time, high-value fund transfers (typically ₹2 lakh and above)
IMPS (Immediate Payment Service)	Instant interbank transfer, available 24×7

Cyber Safety in Banking

- Never share OTPs, PINs, or passwords with anyone, including callers claiming to be from the bank.
- Verify links and senders before clicking; banks never ask for confidential details over phone, SMS or email.
- Use official banking apps and secure networks for financial transactions.

Other Key Financial & Regulatory Institutions

- **SEBI** (Securities and Exchange Board of India): regulates the stock markets and protects investors-12 Apr 1988.
- **IRDAI** (Insurance Regulatory and Development Authority of India): regulates the insurance sector-1999, Headquarter: Hyderabad.
- **NABARD** (National Bank for Agriculture and Rural Development): supports agricultural and rural credit-12 Jul 1982
- **LIC** (Life Insurance Corporation of India): India's largest life-insurance provider, a public sector enterprise.-1956

Types of Taxes

- **Direct Tax:** paid directly by an individual or entity to the government — for example, Income Tax.
- **Indirect Tax:** collected via an intermediary on goods and services — for example, GST.
- **GST (Goods and Services Tax):** a unified indirect tax introduced on 1 July 2017, replacing several earlier indirect taxes across India.

Basic Economic Terms — Quick Glossary

- **GDP (Gross Domestic Product):** the total value of all goods and services produced within a country in a given period.
- **Inflation:** a general rise in prices over time, reducing the purchasing power of money.
- **Fiscal Deficit:** the gap between the government's total expenditure and its total revenue (excluding borrowings).
- **Subsidy:** financial support given by the government to reduce the cost of a good or service for consumers or producers.
- **Disinvestment:** the sale of government-owned stakes in public sector enterprises.

5. Geography of India & Jammu and Kashmir

This section covers India's major physical features — rivers, lakes, deserts — along with its states, Union Territories and capitals, the geography of Jammu & Kashmir, and important protected areas.

Major Rivers of India

River	Origin	Notable Facts
Ganga	Gangotri Glacier, Uttarakhand	Longest river in India; considered sacred
Yamuna	Yamunotri Glacier	Largest tributary of the Ganga
Brahmaputra	Tibet (as Yarlung Tsangpo)	Enters India via Arunachal Pradesh, flows through Assam
Indus	Tibet, flows through Ladakh	Indus Valley Civilisation developed along its banks
Godavari	Trimbakeshwar, Maharashtra	Largest river in peninsular India; called 'Dakshin Ganga'

Other major rivers include the Krishna, Narmada, Tapi and Kaveri.

Major Lakes and Deserts

- **Lakes:** Wular Lake (J&K — largest freshwater lake in India), Dal Lake (Srinagar), Chilika Lake (Odisha — largest coastal lagoon in India), Vembanad Lake (Kerala), Sambhar Lake (Rajasthan — largest saltwater lake).
- **Deserts:** Thar Desert (Rajasthan — largest desert in India); cold desert regions in Ladakh.

States, Union Territories & Capitals

India comprises 28 states and 8 Union Territories (as of 2026), unchanged since the merger of Daman and Diu with Dadra and Nagar Haveli in January 2020.

State	Capital	State	Capital
Andhra Pradesh	Amaravati	Maharashtra	Mumbai
Arunachal Pradesh	Itanagar	Manipur	Imphal
Assam	Dispur	Meghalaya	Shillong
Bihar	Patna	Mizoram	Aizawl
Chhattisgarh	Raipur	Nagaland	Kohima
Goa	Panaji	Odisha	Bhubaneswar
Gujarat	Gandhinagar	Punjab	Chandigarh
Haryana	Chandigarh	Rajasthan	Jaipur
Himachal Pradesh	Shimla	Sikkim	Gangtok
Jharkhand	Ranchi	Tamil Nadu	Chennai
Karnataka	Bengaluru	Telangana	Hyderabad
Kerala	Thiruvananthapuram	Tripura	Agartala
Madhya Pradesh	Bhopal	Uttar Pradesh	Lucknow
		Uttarakhand	Dehradun
		West Bengal	Kolkata
Union Territory		Capital / Administrative Centre	
Andaman & Nicobar Islands		Port Blair	
Chandigarh		Chandigarh	
Dadra & Nagar Haveli and Daman & Diu		Daman	
Delhi (NCT)		New Delhi	
Jammu & Kashmir		Srinagar (summer) / Jammu (winter)	

State	Capital	State	Capital
Ladakh		Leh	
Lakshadweep		Kavaratti	
Puducherry		Puducherry	

Geography of Jammu & Kashmir

- **Major rivers:** Jhelum, Chenab, and the Indus (through Ladakh).
- **Major lakes:** Dal Lake and Wular Lake (Kashmir Valley), Pangong Tso (Ladakh).
- **Capitals:** Srinagar (summer capital) and Jammu (winter capital).

National Parks & Wildlife Sanctuaries

Park	Location	Known For
Jim Corbett National Park	Uttarakhand	First national park in India; tigers
Kaziranga National Park	Assam	One-horned rhinoceros
Sundarbans National Park	West Bengal	Mangroves; Royal Bengal Tiger
Gir National Park	Gujarat	Asiatic lions
Ranthambore National Park	Rajasthan	Tigers
Dachigam National Park	Jammu & Kashmir	Hangul (Kashmir stag)
Kishtwar National Park	Jammu & Kashmir	Snow leopard, Himalayan brown bear

Physical Divisions of India

- The Himalayas — the great northern mountain range, home to the world's highest peaks.
- Northern Plains — the fertile Indo-Gangetic Plains, formed by the Ganga, Yamuna and Brahmaputra river systems.
- Peninsular Plateau — the Deccan Plateau, one of the oldest landmasses in the world.
- Coastal Plains — the Eastern Coastal Plain and Western Coastal Plain.
- Islands — Andaman & Nicobar Islands (Bay of Bengal) and Lakshadweep (Arabian Sea).

India's Neighbouring Countries

Land borders: Pakistan, China, Nepal, Bhutan, Bangladesh, Myanmar and Afghanistan. **Maritime neighbours:** Sri Lanka and the Maldives.

6. Computers & Artificial Intelligence

Basics of Computers

A computer works on the input–process–output cycle: it takes in data, processes it, and produces a result.

- **Hardware:** the physical components — CPU (the 'brain' of the computer), monitor, keyboard, mouse, printer, RAM, and hard disk/SSD.
- **Software:** sets of instructions that tell hardware what to do — System Software (operating systems like Windows, macOS, Linux, Android) and Application Software (word processors, browsers, apps).
- **Input devices:** keyboard, mouse, scanner, microphone, touchscreen.
- **Output devices:** monitor, printer, speaker.

Cyber Safety & Digital Literacy

- Use strong, unique passwords, and enable two-factor authentication where possible.
- Be cautious of phishing emails or messages asking for personal or financial information.
- Think before sharing personal information or photos online; understand privacy settings on social media.
- Report cyberbullying or suspicious online activity to a trusted adult or the appropriate authority.

Basics of Artificial Intelligence (AI)

AI refers to computer systems designed to perform tasks that typically require human intelligence — such as understanding language, recognising images, or making decisions.

- **Applications:** voice assistants, recommendation systems, navigation apps, chatbots, medical-diagnosis support, and self-driving vehicle research.
- **Advantages:** increases efficiency, processes large amounts of data quickly, is available 24/7, and can assist with repetitive or dangerous tasks.
- **Limitations:** can make mistakes or reflect biases present in its training data, lacks genuine human understanding or emotion, and raises questions around privacy and job displacement.
- **Responsible use:** verify AI-generated information before relying on it, be mindful of privacy when sharing data with AI tools, and use AI to support — not replace — one's own learning and judgment.

Generations of Computers

Generation	Period	Key Technology
First	1940s–1950s	Vacuum tubes
Second	1950s–1960s	Transistors
Third	1960s–1970s	Integrated Circuits (ICs)
Fourth	1970s–present	Microprocessors
Fifth	Present onward	Artificial Intelligence-based systems

Internet Basics — Quick Glossary

- **WWW (World Wide Web):** a system of interlinked web pages accessed via the internet.
- **URL:** the web address used to locate a specific page or resource online.
- **Browser:** software used to access and view websites (e.g. Chrome, Firefox, Safari).
- **Search Engine:** a tool that helps find information on the web (e.g. Google, Bing).
- **Wi-Fi:** a technology that allows devices to connect to the internet wirelessly.

7. Quantitative Aptitude (Math)

This section covers simple, calculator-free arithmetic skills that are fair to students of every stream — no algebra or advanced formulas required.

Basic Number Sense

- **Natural numbers:** 1, 2, 3, 4, ... (counting numbers)
- **Whole numbers:** 0, 1, 2, 3, ... (natural numbers plus zero)
- **Integers:** ..., -2, -1, 0, 1, 2, ... (whole numbers plus negative numbers)

Percentage

Formula: $\text{Percentage} = (\text{Part} \div \text{Whole}) \times 100$

Example: If a student scores 45 out of 50 marks, $\text{percentage} = (45 \div 50) \times 100 = 90\%$.

Average

Formula: $\text{Average} = \text{Sum of all values} \div \text{Number of values}$

Example: Average of 10, 20 and 30 = $(10 + 20 + 30) \div 3 = 20$.

Profit and Loss

- Profit = Selling Price (SP) – Cost Price (CP), when SP is greater than CP
- Loss = Cost Price (CP) – Selling Price (SP), when CP is greater than SP
- Profit % = $(\text{Profit} \div \text{CP}) \times 100$ Loss % = $(\text{Loss} \div \text{CP}) \times 100$

Example: An item bought for ₹200 is sold for ₹250. Profit = $250 - 200 = ₹50$. Profit % = $(50 \div 200) \times 100 = 25\%$.

Simple Interest

Formula: $\text{Simple Interest (SI)} = (\text{Principal} \times \text{Rate} \times \text{Time}) \div 100$

Example: ₹1,000 is deposited at 5% annual interest for 2 years. $SI = (1000 \times 5 \times 2) \div 100 = ₹100$.

Ratio & Proportion

A ratio compares two quantities of the same kind. It is written as a : b (read 'a is to b').

Example: If a class has 20 boys and 15 girls, the ratio of boys to girls = 20:15, which simplifies to 4:3 (dividing both terms by 5).

8. Health & Nutrition

Communicable Diseases

These spread from person to person, or via vectors such as mosquitoes, and via contaminated water or food.

Disease	Cause / Agent	Mode of Spread
Common cold / Influenza	Virus	Air droplets
Tuberculosis (TB)	Bacteria	Air droplets
Malaria	Parasite (Anopheles mosquito)	Mosquito bite
Dengue	Virus (Aedes mosquito)	Mosquito bite
COVID-19	Virus	Air droplets / contact
Cholera / Typhoid	Bacteria	Contaminated water or food

Non-Communicable Diseases

These are not spread from person to person, and are often linked to lifestyle, diet and genetics rather than infection — for example diabetes, hypertension (high blood pressure), heart disease, and obesity.

Water-Borne Diseases

Caused by contaminated water, these include cholera, typhoid, hepatitis A, and diarrhoeal diseases. Prevention relies on safe drinking water, proper sanitation, and hand hygiene.

Balanced Diet & Deficiency Diseases

A balanced diet includes carbohydrates, proteins, fats, vitamins, minerals, fibre and water in the right proportions.

Deficiency	Disease / Condition
Vitamin A	Night blindness
Vitamin C	Scurvy
Vitamin D	Rickets (children) / Osteomalacia (adults)
Vitamin B1 (Thiamine)	Beriberi
Iron	Anaemia
Iodine	Goitre
Protein	Kwashiorkor

Human Body — Major Systems

System	Main Function
Digestive System	Breaks down food into nutrients the body can absorb and use
Respiratory System	Enables breathing — the exchange of oxygen and carbon dioxide
Circulatory System	Pumps blood (heart) to carry oxygen and nutrients throughout the body
Nervous System	Controls and coordinates body activities via the brain and nerves
Skeletal System	Provides structure, support and protection for internal organs

Basic First Aid

- **Minor cuts:** clean the wound with water, apply an antiseptic, and cover with a clean bandage.
- **Minor burns:** cool the area under running water for several minutes; do not apply ice directly.
- **Fainting:** lay the person down, raise their legs slightly, and ensure fresh air.
- **Always:** call for adult or medical help promptly for anything beyond a minor injury.

9. Sports

Major National & International Sporting Events

- **Olympic Games:** global multi-sport event; Summer and Winter Olympics each held every 4 years, alternating every 2 years.
- **Asian Games:** multi-sport event for Asian nations, held every 4 years.
- **Commonwealth Games:** multi-sport event for Commonwealth nations, held every 4 years.

Important Tournaments & Trophies (Sport-wise)

Sport	Major Tournaments / Trophies
Cricket	ICC Cricket World Cup, ICC T20 World Cup, Indian Premier League (IPL), Ranji Trophy (domestic, India)
Football	FIFA World Cup, UEFA Champions League, Indian Super League (ISL)
Hockey	Hockey World Cup, Sultan Azlan Shah Cup
Tennis	Grand Slams — Australian Open, French Open, Wimbledon, US Open
Badminton	Thomas Cup (men's team), Uber Cup (women's team), All England Championship
Chess	Chess Olympiad, World Chess Championship

Famous Sportspersons (India) — Illustrative

- **Cricket:** Sachin Tendulkar, Kapil Dev, MS Dhoni, Virat Kohli.
- **Hockey:** Major Dhyan Chand, the 'Hockey Wizard,' who led India to Olympic gold multiple times.
- **Badminton:** Saina Nehwal, P.V. Sindhu — Olympic medallists.
- **Athletics:** Neeraj Chopra — Olympic gold in javelin throw, Tokyo 2020.
- **Wrestling / Boxing:** Sushil Kumar, Mary Kom.

Note: Sporting achievements, records and rankings change frequently — follow recent sports news for the latest updates closer to the quiz.

National Sports Awards (India)

- **Major Dhyan Chand Khel Ratna:** India's highest sporting honour, for the most spectacular and outstanding performance.
- **Arjuna Award:** for consistently outstanding performance in sports over four years.
- **Dronacharya Award:** for excellence in sports coaching.

India at the Olympics — Landmark Moments

- **1952 Helsinki:** K.D. Jadhav wins bronze in wrestling — India's first individual Olympic medal.
- **2008 Beijing:** Abhinav Bindra wins gold in shooting — India's first individual Olympic gold medal.
- **2020 Tokyo (held in 2021):** Neeraj Chopra wins gold in the javelin throw — India's first Olympic gold in athletics.

10. Current Affairs

Current Affairs tests awareness of recent developments and connects classroom learning to the real world. Because this section changes constantly, treat the content below as a framework, and refresh it using recent newspapers or news sources closer to the date of the quiz.

Key Categories to Follow

- National and international news — major policy decisions, summits, and agreements.
- Awards and honours — Bharat Ratna, Padma Awards, Nobel Prizes, sports awards (Khel Ratna, Arjuna Award).
- Government schemes and appointments — new initiatives and key officials.
- Notable sports and cultural events.

Illustrative 'Who's Who' (India)

Position	Held By (as of mid-2026)
President of India	Droupadi Murmu
Vice-President of India	C. P. Radhakrishnan
Prime Minister of India	Narendra Modi
Speaker, Lok Sabha	Om Birla
Chief Justice of India	Justice Surya Kant
RBI Governor	Sanjay Malhotra
Chief Election Commissioner	Gyanesh Kumar

Note: These positions can change. Re-verify each name closer to the date of the quiz.

Awards — Quick Reference

- **Bharat Ratna:** India's highest civilian award, for exceptional service in any field.
- **Padma Awards:** Padma Vibhushan, Padma Bhushan, and Padma Shri (in decreasing order of distinction), for distinguished contributions in various fields.
- **Nobel Prizes:** awarded in Physics, Chemistry, Physiology or Medicine, Literature, Peace, and Economic Sciences.

Notable Government Schemes (Long-Running, Illustrative)

- **PM-Kisan Samman Nidhi:** income support for farmers.
- **Ayushman Bharat:** health insurance coverage scheme.
- **Swachh Bharat Mission:** sanitation and cleanliness campaign.
- **Digital India:** promoting digital infrastructure and services.
- **Beti Bachao Beti Padhao:** welfare and education of the girl child.

Important National & International Days

Day	Date	Category
Republic Day	26 January	National
International Women's Day	8 March	International
Labour Day	1 May	International
Independence Day	15 August	National
Teachers' Day	5 September	National
Gandhi Jayanti	2 October	National
Children's Day	14 November	National
Human Rights Day	10 December	International

Practice Questions

The following sample questions span all ten topics (two questions per topic) and mirror the style used in the syllabus. Answers are given in the Answer Key on the next page.

Q1. Who led the Dandi March in 1930?

- (A) Jawaharlal Nehru (B) Mahatma Gandhi (C) Subhas Chandra Bose (D) Bal Gangadhar Tilak

Q2. The Jallianwala Bagh Massacre took place in which city?

- (A) Delhi (B) Amritsar (C) Lahore (D) Lucknow

Q3. Which Article of the Constitution is often called the 'heart and soul' of the Constitution?

- (A) Article 14 (B) Article 19 (C) Article 21 (D) Article 32

Q4. Who chaired the Drafting Committee of the Indian Constitution?

- (A) Jawaharlal Nehru (B) Dr. B.R. Ambedkar (C) Sardar Patel (D) Rajendra Prasad

Q5. Which international treaty aims to reduce ozone-depleting substances?

- (A) Kyoto Protocol (B) Montreal Protocol (C) Paris Agreement (D) Basel Convention

Q6. Which of the following is a renewable resource?

- (A) Coal (B) Petroleum (C) Solar energy (D) Natural gas

Q7. What does RTGS stand for?

- (A) Real Time Gross Settlement (B) Rapid Transfer Government Service (C) Regional Transaction General System (D) Reserve Transfer Growth Scheme

Q8. Which of these is India's central bank?

- (A) SBI (B) RBI (C) NABARD (D) IDBI

Q9. Which is the largest desert in India?

- (A) Thar Desert (B) Kutch Desert (C) Ladakh cold desert (D) Rann of Kutch

Q10. Which lake in Jammu & Kashmir is the largest freshwater lake in India?

- (A) Dal Lake (B) Wular Lake (C) Chilika Lake (D) Sambhar Lake

Q11. Which of these is an example of an input device?

- (A) Monitor (B) Printer (C) Keyboard (D) Speaker

Q12. What does 'AI' stand for?

- (A) Automatic Information (B) Artificial Intelligence (C) Applied Interface (D) Advanced Internet

Q13. An item bought for ₹200 is sold for ₹250. What is the profit percentage?

- (A) 20% (B) 25% (C) 30% (D) 50%

Q14. What is the average of 10, 20 and 30?

- (A) 15 (B) 20 (C) 25 (D) 30

Q15. Deficiency of which vitamin causes night blindness?

- (A) Vitamin A (B) Vitamin C (C) Vitamin D (D) Vitamin B1

Q16. Which of these is a water-borne disease?

- (A) Malaria (B) Tuberculosis (C) Typhoid (D) Influenza

Q17. The Thomas Cup is associated with which sport?

- (A) Tennis (B) Badminton (C) Hockey (D) Chess

Q18. Who is known as the 'Hockey Wizard' of India?

- (A) Sardar Singh (B) Dhyan Chand (C) Balbir Singh (D) Sandeep Singh

Q19. As of 2026, who is the Reserve Bank of India Governor?

- (A) Shaktikanta Das (B) Urjit Patel (C) Sanjay Malhotra (D) Raghuram Rajan

Q20. How many states and Union Territories does India currently have in total?

- (A) 28 states, 8 UTs (B) 29 states, 7 UTs (C) 27 states, 9 UTs (D) 28 states, 9 UTs

Answer Key

Q#	Answer	Q#	Answer	Q#	Answer	Q#	Answer
Q1	B	Q6	C	Q11	C	Q16	C
Q2	B	Q7	A	Q12	B	Q17	B
Q3	D	Q8	B	Q13	B	Q18	B
Q4	B	Q9	A	Q14	B	Q19	C
Q5	B	Q10	B	Q15	A	Q20	A

General Knowledge Quiz Syllabus

Classes 9th – 12th | All Streams

Scope: General-awareness level, suitable for any stream (Science, Arts, General) — no stream-specific depth required.

Total: 60 Marks / 60 Questions (1 mark each).

Marks & Question Distribution

#	Topic	Marks	Questions
1	Indian Freedom Struggle (1857–1947)	10	10
2	Political Science: Constitution & Governance	10	10
3	Environmental Studies	10	10
4	Basic Economics & Banking	5	5
5	Geography of India & J&K	5	5
6	Computers & Artificial Intelligence	5	5
7	Quantitative Aptitude (Math)	5	5
8	Health & Nutrition	4	4
9	Sports	3	3
10	Current Affairs	3	3
	Total	60	60

Detailed Syllabus

1. Indian Freedom Struggle (1857–1947) (10 marks) • Revolt of 1857 — causes, leaders, consequences • Formation of the Indian National Congress • Partition of Bengal • Home Rule Movement • Jallianwala Bagh Massacre • Non-Cooperation Movement • Civil Disobedience Movement & Dandi March • Quit India Movement	2. Political Science: Constitution & Governance (10 marks) • Fundamental Rights and Duties • Preamble to the Constitution • Landmark Articles (e.g. 14, 19, 21, 32, 44, 356, 370) • Government structure: Parliament, President, PM, Judiciary • Major national & regional political parties
3. Environmental Studies (10 marks) • Greenhouse Effect & Global Warming; Kyoto Protocol • Ozone Layer Depletion; Montreal Protocol • Biodiversity — importance & threats • Renewable and Non-Renewable Resources	4. Basic Economics & Banking (5 marks) • Functions and types of banks in India • RBI — history, objectives, functions • Types of bank accounts • Cheques, passbooks, debit/credit cards • UPI, NEFT, RTGS, IMPS, mobile banking
5. Geography of India & J&K (5 marks) • Major rivers, lakes, deserts of India • States, UTs, capitals, important cities • Major rivers & lakes of J&K • National parks & wildlife sanctuaries (India & J&K)	6. Computers & Artificial Intelligence (5 marks) • Computer basics — hardware, software, I/O devices • Cyber safety, digital literacy, online fraud prevention • AI basics — applications, advantages, limitations, responsible use
7. Quantitative Aptitude (Math) (5 marks) • Basic number sense (integers, whole & natural numbers) • Percentage • Average • Profit and Loss	8. Health & Nutrition (4 marks) • Communicable & Non-Communicable Diseases • Water-borne Diseases • Balanced Diet & Deficiency Diseases
9. Sports (3 marks) • Major national & international sports • Important tournaments and trophies • Famous sportspersons	10. Current Affairs (3 marks) • National & international news (last 6–12 months) • Awards and honours • Government schemes and appointments • Notable sports and other current events

